



Evaluation of pathological alpha-synuclein in skin biopsies as potential biomarker for Parkinson's disease

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Summary

Biomarkers to detect and quantify pathological alpha-synuclein (aSyn) in accessible samples from Parkinson's disease (PD) patients are urgently needed. We have developed a ligand-binding assay (LBA) for aSyn phosphorylated at serine 129 (pS129). Here, we measured pS129 aSyn in skin biopsies from prodromal, sporadic PD, and control individuals from the PPMI cohort. For this measurement skin biopsy homogenates from 99 individuals were used, that were left over from project 262. Our primary aim of this study was to determine the diagnostic accuracy of the pS129 aSyn LBA for skin biopsy homogenates.

Method

pS129-aSyn LBA

The assay was run on the Singulex Erenna platform (Merck, Darmstadt, Germany) using a capture antibody selective for pS129 aSyn and a detection antibody selective for aSyn. Capture and detection antibodies were labelled using Capture/Detection Labeling Kits according to manufacturer's instructions (Merck, Darmstadt, Germany). Magnetic beads coupled with capture antibodies were prepared in Coated Bead Buffer at 10 mg/mL. Detection antibody was adjusted to 1 mg/mL during labeling with fluorescent dye. Skin lysate samples (1:5 dilution) and calibrators were diluted in Standard Diluent (Merck, Darmstadt, Germany) and run in two technical replicates. One hundred μ L of diluted sample or calibrator was added per well of a 96-well plate (Corning, Reynosa, Mexico). Capture beads were diluted 1:200 in Assay/Discovery buffer (Merck, Darmstadt, Germany) and 100 μ L/well was added (5 μ g beads/well). The plate was incubated for 2 h at 25 °C in a shaker. Beads were then washed with Erenna System buffer (Merck, Darmstadt, Germany) and resuspended with 20 μ L/well of 550 ng/mL detection antibody in Assay/Discovery buffer. The plate was incubated for 1 h at 25 °C in a shaker. After washing, the beads were resuspended and transferred into a new 96-well plate. The wash buffer was then removed from beads in a final aspiration step. A total of 9.5 μ L/well of Elution buffer B (Merck, Darmstadt, Germany) was added to the beads and incubated for 20 min at 25 °C in a shaker. The eluate was magnetically separated from the beads, transferred into a 384-well plate (Corning, Reynosa, Mexico) and neutralized immediately with 9.5 μ L/well of Buffer D (Merck, Darmstadt, Germany).





After heat-sealing, the plate was analyzed using the Singulex Erenna instrument. The dilution corrected limit of detection (LOD) for skin lysates was estimated as 7.5 pg/mL. Plate quality control samples (skin homogenate pools) showed less than $\pm 11\%$ deviation from the mean across 4 plates.

Table 1: Overview of the terms used in the results file.

TESTNAME	Description
LBA (rep 1-2)	Ligand-binding assay (LBA) quantification of pS129 aSyn in pg/mL for 2 technical replicates.

About the Authors

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